UNIT 5 CLUSTERING

In **clustering analysis**, the **type of data** determines how similarity or distance between objects is measured.
Different attribute types require different methods for clustering.

The major **types of data in clustering analysis** are:

1. **Interval-scaled variables**
2. **Binary variables**
3. **Nominal variables**
4. **Ordinal variables**
5. **Ratio-scaled variables**
6. **Mixed-type variables**

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1. Interval-Scaled Variables

These are **continuous numerical values** measured on an equal scale.

Examples:

- Age
- Height
- Temperature
- Salary

Example dataset:

Object	Age	Salary
A	20	3000
B	25	5000

To measure similarity, we use **Euclidean distance**:

$$d(i,j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2}$$

These variables are often normalized before clustering because attributes may have different ranges.

Example normalization:

$$z = \frac{x - \mu}{\sigma}$$

$$x$$

$$\mu$$

$$\sigma$$

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

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2. Binary Variables

Binary variables have only **two possible states**:

- 0 or 1
- Yes or No
- True or False

Examples:

- Gender (Male/Female)
- Purchased (Yes/No)

Example:

Object	**Purchased**
A	Yes
B	No

Binary variables are of two types:

(a) Symmetric Binary Variables

Both outcomes are equally important.

Example:

- Gender: Male/Female

Distance can be measured by **simple matching coefficient**:

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

Where:

- **q** = both 1
- **t** = both 0
- **r,s** = mismatches

(b) Asymmetric Binary Variables

One outcome is more important than the other.

Example:

- Disease present (1) / absent (0)

Use **Jaccard coefficient**:

$$d(i,j) = \frac{r+s}{q+r+s}$$

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3. Nominal Variables

Nominal variables represent **categories with no natural order**.

Examples:

- Color = Red, Blue, Green
- Department = Sales, HR, IT

Example:

Object	Color
A	Red
B	Blue

Similarity is measured by:

- Assigning numbers to categories
- Using **matching methods**

Distance:

$$d(i,j) = \frac{p-m}{p}$$

Where:

- **p** = total attributes
- **m** = matches

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4. Ordinal Variables

Ordinal variables have a **meaningful order**, but the differences are not measurable.

Examples:

- Rank: 1st, 2nd, 3rd
- Satisfaction: Low, Medium, High

Example:

Object	Satisfaction
A	High
B	Medium

These are converted into ranks:

Low = 1
Medium = 2
High = 3

Then normalized:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Where:

- r_{if}= rank
- M_f= maximum rank

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5. Ratio-Scaled Variables

Ratio variables are numerical variables with a **true zero point**.

Examples:

- Weight
- Distance
- Income

These can be treated like interval variables.

Sometimes logarithmic transformation is used:

$$y = \log(x)$$

This helps when data has large ranges.

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6. Mixed-Type Variables

Real-world datasets often contain:

- Numerical attributes
- Binary attributes
- Nominal attributes
- Ordinal attributes

Example customer data:

Age	Gender	Income Level	Purchased
25	Male	High	Yes

For mixed data, use **combined dissimilarity measures**, such as **Gower coefficient**.

General formula:

$$d(i,j) = \frac{\sum \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum \delta_{ij}^{(f)}}$$

This combines distances for different attribute types.

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7. Summary Table

Data Type	Example	Distance Measure
Interval	Age, Salary	Euclidean Distance
Binary	Yes/No	Jaccard / Matching
Nominal	Color	Matching Coefficient
Ordinal	Low/Medium/High	Rank-Based Distance
Ratio	Weight, Income	Euclidean / Log Transform

Data Type	Example	Distance Measure
Mixed	Customer Data	Gower Distance

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8. Importance in Clustering

Choosing the correct data type is important because:

- Different data types require different similarity measures
- Wrong distance measures produce poor clusters
- Proper preprocessing improves clustering accuracy

For example:

- **K-Means** works well with **interval data**
- **Hierarchical clustering** can handle **mixed data**
- **DBSCAN** works with **numerical distance-based data**

PARTITIONING METHODS AND HIERARCHICAL METHODS

1. Partitioning Methods

Partitioning methods divide the dataset into **K non-overlapping clusters**, where:

- Each object belongs to exactly **one cluster**
- Number of clusters **K is predefined**

The goal is to optimize cluster quality by minimizing intra-cluster distance and maximizing inter-cluster distance.

Common Partitioning Algorithms

(a) K-Means Clustering

- Divides data into **K clusters**
- Each cluster is represented by the **mean (centroid)**

It minimizes:

$$J = \sum_{j=1}^{K} \sum_{i=1}^{n} \| x_i^{(j)} - c_j \|^2$$

Where:

- c_j is centroid of cluster **j**
- x_i is a data point

Steps:

1. Choose **K**
2. Select **K centroids**
3. Assign points to nearest centroid
4. Recalculate centroids
5. Repeat until convergence

Example:

If data points are:

$$2,\ 4,\ 10,\ 12$$

and **K = 2**, clusters become:

- Cluster 1: {2,4}
- Cluster 2: {10,12}

(b) K-Medoids Clustering

Similar to K-Means, but:

- Uses **actual data points** as centers (**medoids**)
- Less sensitive to outliers

Instead of mean, it selects the most centrally located object.

Example:

If cluster is:

$$\{2, 4, 10\}$$

The medoid is the point with minimum total distance to others.

Advantages of Partitioning Methods

- Simple and efficient
- Suitable for large datasets

- Fast convergence

Disadvantages

- Need to specify **K**
- Sensitive to initial cluster centers
- Not good for irregular cluster shapes

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2. Hierarchical Methods

Hierarchical clustering creates a **hierarchy of clusters** represented as a **tree structure (dendrogram)**.

It does **not require K initially**.

There are two approaches:

(a) Agglomerative Method (Bottom-Up)

- Start with each object as one cluster
- Merge nearest clusters repeatedly
- Continue until one cluster remains

Process:

$$\{A\}, \{B\}, \{C\}, \{D\}$$

After merging:

$$\{A, B\}, \{C, D\}$$

Then:

$$\{A, B, C, D\}$$

(b) Divisive Method (Top-Down)

- Start with all objects in one cluster
- Split recursively into smaller clusters

Example:

$$\{A, B, C, D\}$$

Split into:

$$\{A, B\}, \{C, D\}$$

Then split further.

Distance Measures in Hierarchical Clustering

Single Linkage

Minimum distance between two clusters:

$$D(A, B) = \min(d(x, y))$$

Complete Linkage

Maximum distance:

$$D(A, B) = \max(d(x, y))$$

Average Linkage

Average distance:

$$D(A, B) = \frac{1}{|A||B|} \sum d(x, y)$$

Dendrogram Example

```
    {A,B,C,D}
    /     \
  {A,B}   {C,D}
  / \     / \
 A   B   C   D
```

This tree shows how clusters are merged.

Advantages of Hierarchical Methods

- No need to specify number of clusters
- Dendrogram gives visual representation
- Easy to interpret

Disadvantages

- Computationally expensive
- Difficult for very large datasets
- Cannot undo previous merges

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3. Difference Between Partitioning and Hierarchical Methods

Feature	Partitioning	Hierarchical
Number of clusters	Predefined	Not predefined
Structure	Flat clusters	Tree hierarchy
Speed	Faster	Slower
Scalability	Better for large data	Better for small data
Example	K-Means	Agglomerative

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4. Applications in Data Mining

Both methods are used for:

- **Customer segmentation**
- **Pattern recognition**
- **Market analysis**
- **Image segmentation**
- **Document clustering**

Example in business:

Customers can be clustered into:

- High-value customers
- Medium-value customers
- Low-value customers

This helps in decision-making and marketing.

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5. Summary

Partitioning Methods:

- Divide data into **K clusters**
- Example: **K-Means, K-Medoids**

Hierarchical Methods:

- Build cluster hierarchy